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Editors  
Journal of Public Economics

Dear Professors Hendren and Kopczuk,

We are submitting “Private Annuities and Social Security Cuts: An Order-Independent Decomposition of the Annuity Puzzle” for consideration at the *Journal of Public Economics*.

Yaari (1965) proved that rational consumers should fully annuitize under actuarially fair pricing, yet voluntary ownership among US retirees is only a few percent. Six decades of research proposed competing “dominant” explanations—bequests, correlated health and mortality risk, pricing loads—but the sequential decompositions used to adjudicate them are order-dependent, so the field’s answer has depended on which channel is switched on first. Pashchenko (2013, *JPubE*) came closest to a unified accounting but still predicted participation near 20%.

The paper nests nine rational and preference channels in one calibrated lifecycle model and attributes the annuity-demand gap with exact Shapley values over all  $2^9 = 512$  channel subsets, so the attribution is order-independent by construction. The ranking is unambiguous. Pricing loads are the dominant suppressor of demand (31.4 pp), with survival pessimism (10.9 pp) and the combined medical-expenditure and health-mortality channel (9.4 pp) as the co-leading second tier. Bequest motives, the single most-cited explanation in the literature, rank mid-pack (5.8 pp). The order-dependence of the prior decompositions was itself a source of their disagreement, and the Shapley adjudicates a twenty-year argument rather than adding another entry to it.

A second contribution is methodological. The predicted ownership *level* is knife-edged in risk aversion—it moves from 0.0% to 21.9% as  $\gamma$  ranges over  $[2.0, 3.0]$ —because the extensive margin sits on a minimum-purchase threshold that the population crosses discontinuously. The same fragility plausibly explains why comparable models have predicted ownership anywhere from a few percent to full annuitization. The channel ranking, by contrast, is stable across risk aversion, across wealth quartiles, and across the discrete-versus-continuous demand statistic. In calibrated annuity models, the ranking and the policy comparative statics, not the predicted level, are the objects on which to stake claims.

The resolution reframes the puzzle as distributional, which is where the public-finance stakes lie. The same pricing load is the binding margin across the wealth distribution: at the baseline money’s-worth ratio, immediate annuitization is value-destroying for all but the wealthiest, so predicted ownership rises with wealth and concentrates in the top quartile (33.5%, against 0.0% in the bottom). Observed lifetime-annuity ownership also rises with wealth, but it is interior in the middle of the distribution, where the single-product model predicts none—a gap we attribute to better-priced employer and group annuities the model omits. The policy counterfactuals inherit this structure. A 22% Social Security benefit cut, the projected consequence of trust-fund exhaustion in late 2032, raises private annuity demand mainly in the top quartile (from 33.5% to 45.4%), while barely moving the lower-wealth households the cut most exposes. Privatized longevity insurance does not backfill a public benefit cut where it bites hardest; the lever that would is group or public-option pricing, not individual annuitization. The model’s cross-sectional predictions match the HRS data used to discipline it (5 of 7 channel-implied signs reproduced), and the structural prediction (7.9%)

sits modestly above the two HRS ownership measures reported in parallel (3.64% on the lifetime-contract indicator, 6.06% on the any-annuity income proxy), neither of which enters the calibration.

Two exploratory behavioral channels, source-dependent utility and a narrow-framing at-purchase penalty, are reported separately. Calibrated to literature magnitudes rather than US moments, they move the prediction substantially but offset, and do not disturb the structural ranking. We are explicit throughout that this extension is illustrative rather than identified.

The paper engages the journal's core interests in Social Security, retirement saving, insurance-market frictions, and the design of public and employer-sponsored retirement institutions, and it engages directly with Pashchenko (2013) and related work in *JPubE*. Replication code in Julia and all calibration data are publicly available. The manuscript has not been submitted elsewhere and is not under consideration at another journal.

Sincerely,

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